

EDUCATION

Dalhousie University Ph.D. in Mathematics, Supervisor: Neil J. Ross Thesis: <i>Ongoing</i>	Halifax, NS, Canada 2021–Present
University of Waterloo M.Math. in Computer Science, Supervisors: Arie Gurfinkel and Richard Trefler Thesis: Smart Contract Analysis Through Communication Abstractions	Waterloo, ON, Canada 2019–2021
University of Waterloo B.Math. in Computer Science Honours with Distinction, Co-operative Education Program	Waterloo, ON, Canada 2014–2019

WORK EXPERIENCE

Uber Advanced Technology Group Software Engineer Intern	Pittsburgh, PA, USA May 2018–Dec. 2018
DST Systems, Inc. R&D Developer	Toronto, ON, Canada May 2016–April 2017
AGF Management Limited Junior Web Developer	Toronto, ON, Canada May 2015–Aug. 2015

PUBLICATIONS

- [1] F. Bayeh, L. Dinh, D. Lee, and S. **Wesley**, “Bridging the quantum divide: A learning-centric quantum hackathon for underrepresented students”, in *2025 IEEE International Conference on Quantum Computing and Engineering*, (submitted), IEEE, 2026.
- [2] L. Bu, S. Faridi, I. M. Hewalage, T. Holleben, H. Mahmood, D. Veer, K. Wang, and S. **Wesley**, “When are Morse resolutions polyhedral?”, *Algebra Colloquium*, 2026, (submitted).
- [3] W. Schober and S. **Wesley**, “A complete equational theory for quantum circuits with generalized control”, in *23rd International Conference on Quantum Physics and Logic*, ser. EPTCS, (to appear), Open Publishing Association, 2026.
- [4] N. J. Ross and S. **Wesley**, “Cutoff theorems for the equivalence of parameterized quantum circuits”, in *50th International Symposium on Mathematical Foundations of Computer Science*, ser. LIPIcs, vol. 345, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, 85:1–85:19.
- [5] M. Amy, N. J. Ross, and S. **Wesley**, “A sound and complete equational theory for 3-qubit Toffoli-Hadamard circuits”, in *21st International Conference on Quantum Physics and Logic*, ser. EPTCS, vol. 406, Open Publishing Association, 2024, pp. 1–43.
- [6] S. **Wesley**, “LinguaQuanta: Towards a quantum transpiler between OpenQASM and Quipper”, in *Reversible Computation – 16th International Conference*, ser. LNCS, vol. 14680, Springer, 2024, pp. 142–160.

- [7] S. **Wesley**, M. Christakis, J. Navas, R. Treffer, V. Wüstholtz, and A. Gurfinkel, “Inductive predicate synthesis modulo programs”, in *38th European Conference on Object-Oriented Programming, ECOOP 2024*, ser. LIPIcs, vol. 313, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2024, 43:1–43:30.
- [8] S. **Wesley**, M. Christakis, J. Navas, R. Treffer, V. Wüstholtz, and A. Gurfinkel, “Verifying Solidity smart contracts via communication abstraction in SmartACE”, in *Verification, Model Checking, and Abstract Interpretation – 23rd International Conference*, ser. LNCS, vol. 13182, Springer, 2022, pp. 425–449.
- [9] S. **Wesley**, M. Christakis, J. Navas, R. Treffer, V. Wüstholtz, and A. Gurfinkel, “Compositional verification of smart contracts through communication abstraction”, in *Static Analysis – 28th International Symposium (SAS)*, ser. LNCS, vol. 12913, Springer, 2021, pp. 429–452.
- [10] B. N. Memon, X. C. Lin, A. Mufti, A. S. **Wesley**, T. Brecht, K. Salem, B. Wong, and B. Cassell, “RaMP: A lightweight RDMA abstraction for loosely coupled applications”, in *10th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud 18)*, USENIX Association, 2018.

PRESENTATIONS

Contributed Talks (Conferences/Meetings/Workshops)

- *Parameterized Quantum Circuit Semantics Through Enriched Categories*, 33rd Foundational Methods in Computer Science Workshop (FMCS), Antigonish, Canada, 2026
- *Parameterized Quantum Circuit Semantics Through Enriched Categories*, Canadian Mathematical Society (CMS) Summer Meeting: Category Theory Session, Saint John, Canada, 2026
- *Generalized Quantum Conditional Statements*, Science Atlantic: Mathematics, Statistics, and Computer Science (MSCS), Sydney, Canada, 2025
- *Cutoff Theorems for the Equivalence of Parameterized Quantum Circuits*, 50th International Symposium on Mathematical Foundations of Computer Science (MFCS), Warsaw, Poland, 2025
- *How to Build a Parameterized Function*, 32nd Foundational Methods in Computer Science Workshop (FMCS), Ottawa, Canada, 2025
- *Verifying the Equivalence of Parameterized Quantum Circuits*, 5th International Workshop on Programming Languages for Quantum Computing (PLanQC), Denver, USA, 2025
- *Verifying and Simplifying Tietze Transformations*, Canadian Mathematical Society (CMS) Winter Meeting: Student Research Session, Richmond, Canada, 2024
- *Verifying and Simplifying Tietze Transformations*, Science Atlantic: Mathematics, Statistics, and Computer Science (MSCS), Wolfville, Canada, 2024
- *Inductive Predicate Synthesis Modulo Programs*, 38th European Conference on Object-Oriented Programming (ECOOP), Vienna, Austria, 2024
- *A Sound and Complete Equational Theory for 3-qubit Toffoli-Hadamard Circuits*, 21st International Conference on Quantum Physics and Logic (QPL), Buenos Aires, Argentina, 2024
- *LinguaQuanta: Towards a Quantum Transpiler Between OpenQASM and Quipper*, 16th Conference on Reversible Computation (RC), Toruń, Poland, 2024
- *Verifying Quantum Circuits in GAP*, 1st Workshop on Quantum Software (WQS), Copenhagen, Denmark, 2024
- *Towards an Algebraic and Geometric Theory of Quantum Circuits*, Canadian Mathematical Society (CMS) Winter Meeting: Student Research Session, Montreal, Canada, 2023
- *Visualizing Qudit Controls Through Sheets*, Science Atlantic: Mathematics, Statistics, and Computer Science (MSCS), Charlottetown, Canada, 2023
- *LinguaQuanta: Initial Results on a Quantum Transpiler Between OpenQASM and Quipper with Categorical Specifications*, 5th International Workshop on Quantum Compilation (IWQC), Paris, France, 2023
- *Compositional Verification of Smart Contracts Through Communication Abstraction*, COVID Time Papers In Person (SPLASH), Auckland, New Zealand, 2022

- *Verifying Solidity Smart Contracts Via Communication Abstraction in SmartACE*, 23rd International Conference on Verification, Model Checking, and Abstract Interpretation, Philadelphia, USA (Online), 2022
- *Compositional Verification of Smart Contracts Through Communication Abstraction*, 28th Static Analysis Symposium, Chicago, USA (Online), 2021
- *SmartACE: Functional Verification of Parameterized Smart Contracts*, 1st International Workshop on Smart Contract Analysis, Los Angeles, USA (Online), 2020

Poster Presentations (Conferences/Meetings/Workshops)

- *Enriched Categories for Parameterized Quantum Circuit Semantics*, Young Researchers Forum (YRF) @ 50th International Symposium on Mathematical Foundations of Computer Science (MFCS), Warsaw, Poland, 2025
- *Enriched Categories for Parameterized Circuit Semantics*, 22nd International Conference on Quantum Physics and Logic (QPL), Varna, Bulgaria, 2025
- *On Quantum Circuits Enacting the E_8 Weyl Group*, Canadian Mathematical Society (CMS) Winter Meeting, Montreal, Canada, 2023
- *Automated Smart Contract Analysis*, Blockchain and Security Workshop, Waterloo, 2019
- *Automated Smart Contract Analysis*, Cybersecurity and Privacy Institute Annual Conference, Waterloo, 2019

Miscellaneous Talks

- *Towards Free Control Functors*, Atlantic Category Theory Seminar, Dalhousie University, Halifax, Winter 2026
- *From Monoidal Generators to Control Functors*, Dalhousie Graduate Seminar, Halifax, Winter 2026
- *Generalized Quantum Conditional Statements*, Dalhousie Graduate Seminar, Halifax, Fall 2025
- *How to Build a Parameterized Map*, Atlantic Category Theory Seminar, Dalhousie University, Halifax, Winter 2025
- *Verifying the Equivalence of Parameterized Quantum Circuits*, Dalhousie Graduate Seminar, Halifax, Winter 2025
- *Symbolic and Algebraic Computation (A Tutorial)*, Dalhousie Graduate Seminar, Halifax, Fall 2024
- *Reversible Circuits and Where to Find Them*, Dalhousie Graduate Seminar, Halifax, Winter 2024
- *Visualizing Qudit Controls Through Sheets*, Dalhousie Graduate Seminar, Halifax, Fall 2023
- *Similar Shapes and Topological Magic*, Nova Scotia Math Circles, Halifax, 2022

PROGRAM COMMITTEE EXPERIENCE

- The Sixth International Workshop on Programming Languages for Quantum Computing 2026

OUTREACH EXPERIENCE

Dalhousie Quantum Hacking a Better Future Lead Organizer and Content Developer	Halifax, NS, Canada Jan. 2025–Aug. 2025
Nova Scotia Math Circles Presenter and Content Developer	Halifax, NS, Canada Sept. 2021–Aug. 2024

POST-SECONDARY TEACHING ROLES

Instructor

- **Instructor** at Dalhousie University Fall 2025
Differential & Integral Calculus I (MATH1000)
- **Instructor** at Dalhousie University Winter 2024, Fall 2024

Calculus for the Life and Social Sciences (MATH1215)

- **Instructor** at Dalhousie University Summer 2023
Calculus for the Life and Social Sciences (MATH1215 - Online)
- **Tutorial Instructor** at Dalhousie University Winter 2022, Fall 2022, Winter 2023
Calculus for the Life and Social Sciences (MATH1215)

Teaching Assistant

- **Teaching Assistant** at Dalhousie University Summer 2024
Calculus for the Life and Social Sciences (MATH1215)
- **Teaching Assistant** at Dalhousie University Summer 2022
Linear Algebra 1 (MATH1030)
- **Tutor** at Dalhousie University Fall 2021, Summer 2022, & Fall 2023
Mathematics and Statistics Learning Centre
- **Teaching Assistant** at Dalhousie University Fall 2021, Fall 2022, Fall 2023, & Fall 2024
Applied Graph Theory (MATH3330)
- **Teaching Assistant** at University of Waterloo Winter 2020
Computational Audio (CS489)

Marker

- **Marker** at Dalhousie University Summer 2026
Calculus for the Life and Social Sciences (MATH1215)
- **Marker** at Dalhousie University Summer 2026
Differential & Integral Calculus I (MATH1000)
- **Marker** at Dalhousie University Winter 2025
Problems in Geometry (MATH2051)
- **Marker** at Dalhousie University Winter 2025
Linear Algebra (MATH2135)
- **Marker** at Dalhousie University Winter 2024 & Winter 2025
Discrete Structures II (CSCI2113/MATH2113)
- **Exam Marker** at Dalhousie University Fall 2022
Optimization (MATH3300)
- **Marker** at Dalhousie University Winter 2022
Engineering Mathematics II (MATH1290)
- **Marker** at Dalhousie University Fall 2021 & Winter 2022
Engineering Mathematics I (MATH1280)
- **Marker** at University of Waterloo Winter 2021
Elementary Algorithm Design and Data Abstraction (CS136)
- **Marker** at University of Waterloo Fall 2020
Concurrent and Parallel Programming (CS343)
- **Marker** at University of Waterloo Winter 2020 & Spring 2020
Foundations of Sequential Programs (CS241)
- **Marker** at University of Waterloo Fall 2019
Introduction to Computer Programming 1 (CS105)

Private Tutor

- **Dalhousie University**, Measure Theory and Integration (MATH 5010) Fall 2022

PROFESSIONAL DEVELOPMENT

Atlantic Association for Research in Mathematical Sciences Atlantic Topological Quantum Field Theory (TQFT) Spring School	Halifax, NS, Canada 2026
Atlantic Association for Research in Mathematical Sciences Atlantic Topological Quantum Field Theory (TQFT) Spring School	Fredericton, NB, Canada 2025
Canadian Mathematical Society Mathematical Modelling of Traffic Flow (Mini-Course)	Richmond, BC, Canada 2024
Atlantic Association for Research in Mathematical Sciences Atlantic Topological Quantum Field Theory (TQFT) Spring School	St. John's, NL, Canada 2024
Canadian Mathematical Society Data-Driven Methods for Dynamic Systems (Mini-Course)	Montreal, QB, Canada 2023
Atlantic Association for Research in Mathematical Sciences Atlantic Topological Quantum Field Theory (TQFT) Spring School	Wolfville, NS, Canada 2023

SCHOLARSHIPS, GRANTS AND AWARDS

• AARMS Junior Researcher Travel Support Grant (\$800)	2026
• Dalhousie Mathematics and Statistics Graduate Student Society Pilot Travel Grant Program (\$250)	2026
• Canadian Mathematical Society (CMS) Student Travel Grant (\$200)	2026
• Mary Margeret Werner Graduate Scholarship (\$8000)	2026
• AARMS Junior Researcher Travel Support Grant (\$800)	2025
• Programming Languages for Quantum Computing (PLanQC) Student Travel Grant (\$450)	2025
• CISE-Atlantic Catalyze and Facilitate Sponsorship Program Grant (\$2250)	2025
• Canadian Mathematical Society (CMS) Student Travel Grant (\$200)	2024
• First Place, Dalhousie Qiskit Fallfest by IBM	2024
• Second Place, Graduate Research Award at Science Atlantic: Mathematics, Statistics, and Computer Science	2024
• AARMS Junior Researcher Travel Support Grant (\$800)	2024
• First Place, Qiskit Halifax 2024 Winter Hackathon by IBM	2024
• Canadian Mathematical Society (CMS) Student Travel Grant (\$200)	2023
• Dalhousie Faculty of Science Supplementary Travel Grant (\$500)	2023
• Dalhousie Faculty of Graduate Studies (FGS) Travel Grant (\$500)	2023
• AARMS Junior Researcher Travel Support Grant (\$800)	2023
• Killam Predoctoral Scholarship Level 2 (\$42,000)	2023
• Nova Scotia Graduate Scholarship (\$60,000)	2021
• Waterloo University Mathematics Domestic Masters Scholarship (\$3000)	2019
• Waterloo University President's International Experience Grant (\$1500)	2018
• Waterloo University President's Research Grant (\$1500)	2017
• Waterloo University President's Scholarship of Distinction (\$2000)	2014
• Michelin Nova Scotia Award for Scholastic Excellence (\$2000)	2014

EXTRACURRICULAR ACTIVITIES

- **Finance Co-Chair** of the CUPE 3912 Dalhousie Strike Committee Oct. 2025
Organization of a financial sub-committee to distribute strike pay to union members.
- **Participant** of the Dalhousie Higher Hilbert Spaces Reading Group Jan. 2025–May 2025
Attended meetings, led discussions on 2-categories, unitary adjoint functors, and H^ -monads.*
- **Student Volunteer** at the 29th Workshop on Logic, Language, Information and Computation Jul. 2023
Facilitated hybrid presentations and participated in logistical planning.
- **VP Internal (Treasurer)** of Dalhousie Math/Stat Graduate Student Association Mar. 2023–Sept. 2024
Participates in monthly board meetings and manages finances. Minute keeper.
- **Participant** of the Dalhousie Topological Quantum Field Theory (TQFT) Reading Group Sept. 2022–Dec. 2022
Attended meetings, coordinated technological requirements, led discussion on Hopf algebras.
- **Member** of the Patrick F. Lett Graduate Student Assistance Bursary Committee May 2022–Dec. 2024
Participated in allocations of the Patrick F. Lett Graduate Student Assistance Bursary each semester.
- **Ph.D. Math. Representative** of Dalhousie Math/Stat Graduate Student Association Sept. 2021–Mar. 2023
Participated in the monthly board meetings and departmental event planning. Minute keeper.
- **Member** of the Undergraduate Academic Plans Committee Sept. 2020–Aug. 2021
Participated in the maintenance and development of undergraduate curriculum.
- **Communications Officer** of University of Waterloo CS Graduate Student Association Jan. 2020–Dec. 2020
Drafted official communications and participated in monthly board meetings.